Wiretap Codes and Lattices Satisfying the Belfiore-Solé Conjecture for Length 48

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Goal

The Belfiore-Solé Conjecture states that the signal to noise ratio for a wiretap code has a maximum for unimodular lattices at y=1. It has already been shown that lattices constructed from doubly even self-dual codes (type II codes) from lengths up to 40 satisfy the conjecture. The goal of this project is to extend this to lengths 48, 56, and 64, focusing on 48 primarily.

Introduction

Consider the scenario where two parties want to send information across some noisy channel by encoding the information in some algebraic lattice code and an unwanted third party attempts to listen in. It is a logical question to ask if there is some encoding method that maximizes noise for the unwanted listener while minimizing noise between the two legal users.

In order to solve this problem Belfiore and Oggier introduced the notion of the secrecy function $\Xi_\Lambda(y) = \frac{\Theta_{\mathbb{Z}^n}(iy)}{\Theta_\Lambda(iy)}$ for a lattice code using coset encoding. If we can maximize Ξ then we can maximize the noise to the eavesdropper.

The Belfiore-Solé Conjecture: The secrecy function of a unimodular lattice attains its maximum at $y = 1$ and it is a global maximum.

It was also shown that in order to prove that a given lattice satisfies the conjecture one only needs to show that for some polynomial $p(\zeta)$, with $\zeta \to \zeta(y)$ and $\Xi = [p(\zeta]^{-1}$, that p is decreasing on the interval $\zeta(y) \in [0, \frac{1}{4}]$.

Methodology

The Θ's that defined the secrecy function are the theta functions of the lattices, we can relate these to the spectra of the associated codes by $\Theta_{\Lambda(C)} = W_C(x \to \vartheta_3(2z), y \to \vartheta_2(2z))$. We also use the fact that every Type II code must satisfy $W_C(x, y) \in C[\psi_8, W_{g_{24}}]$. This lets us claim that any type II code of length 48 must have a spectrum that can be expressed as a combination of powers of the polynomials ψ_8 and $W_{g_{24}}$

We then transform the code's spectra into a lattice spectra by using the transformations of the two generator polynomials in terms of Jacobi theta functions ϑ_2, ϑ_3, and ϑ_4.

$$\psi_8 \mapsto \vartheta_3^8(z) - \vartheta_2^4(z)\vartheta_4^4(z)$$

$$W_{g_{24}} \mapsto \vartheta_3^{24}(z) - 3\vartheta_3^{16}(z)\vartheta_2^4(z)\vartheta_4^4(z) + \frac{3}{8}\vartheta_3^8(z)\vartheta_2^8(z)\vartheta_4^8(z) - \vartheta_2^{12}(z)\vartheta_4^{12}(z)$$

Then we use the fact that $\Theta_{\mathbb{Z}^n} = \vartheta_3^n$ to define the polynomial $p(\zeta)$, $\zeta = \frac{\vartheta_2^4\vartheta_4^4}{\vartheta_3^8}$, with two parameters relating to the number of codewords of a given weight. We then only need to show that $p'(\zeta) < 0$, $\zeta \in [0, \frac{1}{4}]$ in the region of parameter space. Unfortunately in order to be rigorous this must be done mostly by exhaustive search, which is computationally extensive.

Parameterization of $W_{C_{48}}(x, y)$ and Associated $p(\zeta)$

We can parameterize the spectrum of a type II code of length 48 as follows for $\eta, \chi \in \mathbb{R}$.

$$W_{C_{48}}(x, y) = (1 - \eta - \chi)\psi_8^6 + \chi\psi_8^3 W_{g_{24}} + \eta W_{g_{24}}^2$$

This can be expanded and simplified to give a system of inequalities in terms of η and χ. Then it can be related to the canonical form of the spectra to get the relations $84 - 84\eta - 42\chi = A_4$ and $2946 - 1428\eta - 1596\chi = A_8$.

Where A_4 and A_8 are the number of words in the code of length 4 and 8.

Applying the transformation to the lattice theta series and dividing by the theta series for $\mathbb{Z}^{48}$ gives

$$p(\zeta) = A_4\left[-\frac{1}{21}\zeta + \frac{841}{7056}\zeta^2 - \frac{35}{336}\zeta^3 + \frac{233}{2688}\zeta^4 - \frac{1}{16}\zeta^5\right] + A_8\left[\frac{1}{441}\zeta^2 - \frac{1}{168}\zeta^3 + \frac{1}{256}\zeta^4\right] + 1 - 2\zeta - \frac{995}{588}\zeta^2 + \frac{44}{7}\zeta^3 - \frac{485}{128}\zeta^4 - \frac{3}{4}\zeta^5 + \zeta^6$$

Which has derivative

$$p'(\zeta) = A_4\left[-\frac{1}{21} + \frac{841}{3528}\zeta - \frac{35}{112}\zeta^2 + \frac{233}{672}\zeta^3 - \frac{5}{16}\zeta^4\right] + A_8\left[\frac{2}{441}\zeta - \frac{1}{56}\zeta^2 + \frac{1}{64}\zeta^3\right] - 2 - \frac{995}{294}\zeta + \frac{132}{7}\zeta^2 - \frac{485}{32}\zeta^3 - \frac{15}{4}\zeta^4 + 6\zeta^5$$

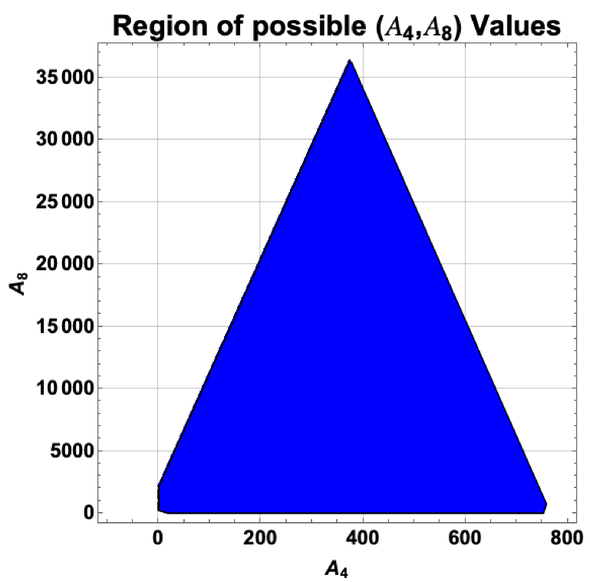


Figure 1:Region of possible A_4 and A_8 values. It is important to note that this graph is over $\mathbb{R}^2$ and A_4 and A_8 must be integer valued.

Based on figure 2 there are 14,381,890 integral points that satisfy the relations.

Reducing the Parameter Space

We can look at the case where $p'(\zeta)$ has a root on the boundary $\zeta \in [0, \frac{1}{4}]$ to obtain two lines $A_4 = -42$ and $\frac{3455742}{473} + \frac{6067}{473}A_4 = A_8$, which split $\mathbb{R}^2$ into four sections.

Looking at a graph of the polynomial one can see that below the line $p'(\frac{1}{4}) = 0$ there are no roots in the interval $[0, 1/4]$ and $p'(\zeta) < 0$, above that line there is a root within the interval close to the $1/4$ boundary and the conjecture is not satisfied. To the right of the vertical line $A_4 = -42$ the polynomial p' will have a negative y-intercept and will only be able to cross the x-axis if A_4 and A_8 are above the second line. To the left of the vertical line the polynomial p' will have a positive y-intercept and will not satisfy the conjecture

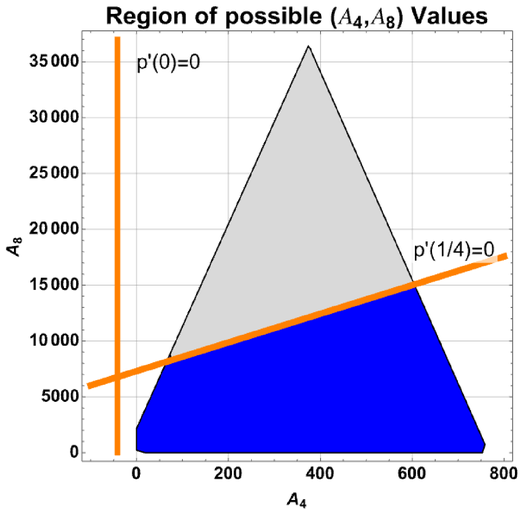


Figure 2:Reduced region of possible A_4 and A_8 values with 7,812,011 points to check

Results Summary and Outlook

We managed to reduce the parameter space for lattices arising from length 48 type II codes satisfying the BS-conjecture. There are roughly 7.8 million parameter pairs that give a valid code and could satisfy the Belfiore-Solé conjecture. However an extensive search is still in progress. Results on the boundary of the region have all confirmed that the blue region satisfies the conjecture. Meaning so far no counter examples in the region below $p'(1/4) = 0$ have been found

The next step is to finish the search in the rest of the region and then to obtain the polynomials for length 56 and 64 and begin to rule out regions of the parameter space that do not satisfy the conjecture.

References

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